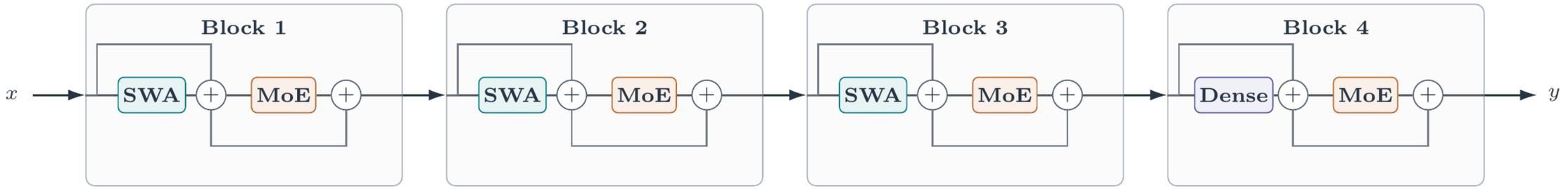
Existing Mellum2 MoE model

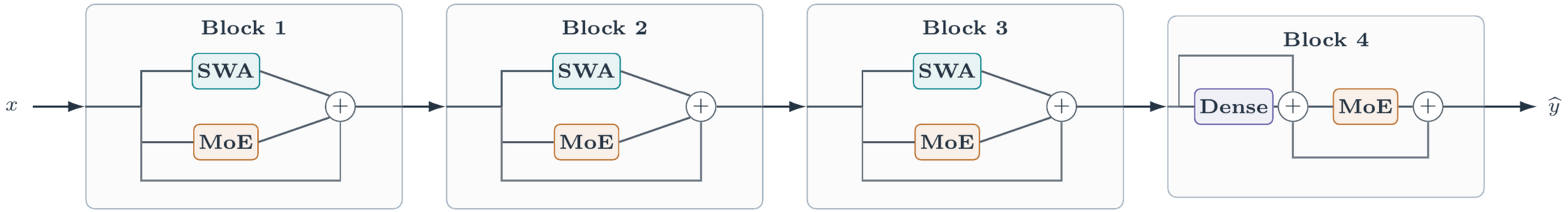
12B params, 2.5B active, 28 decoders, 3 SWA : 1 dense interleaved attention

Existing model: sequential attention → MoE



Rearrange the connections; reuse attention, router, and expert weights.

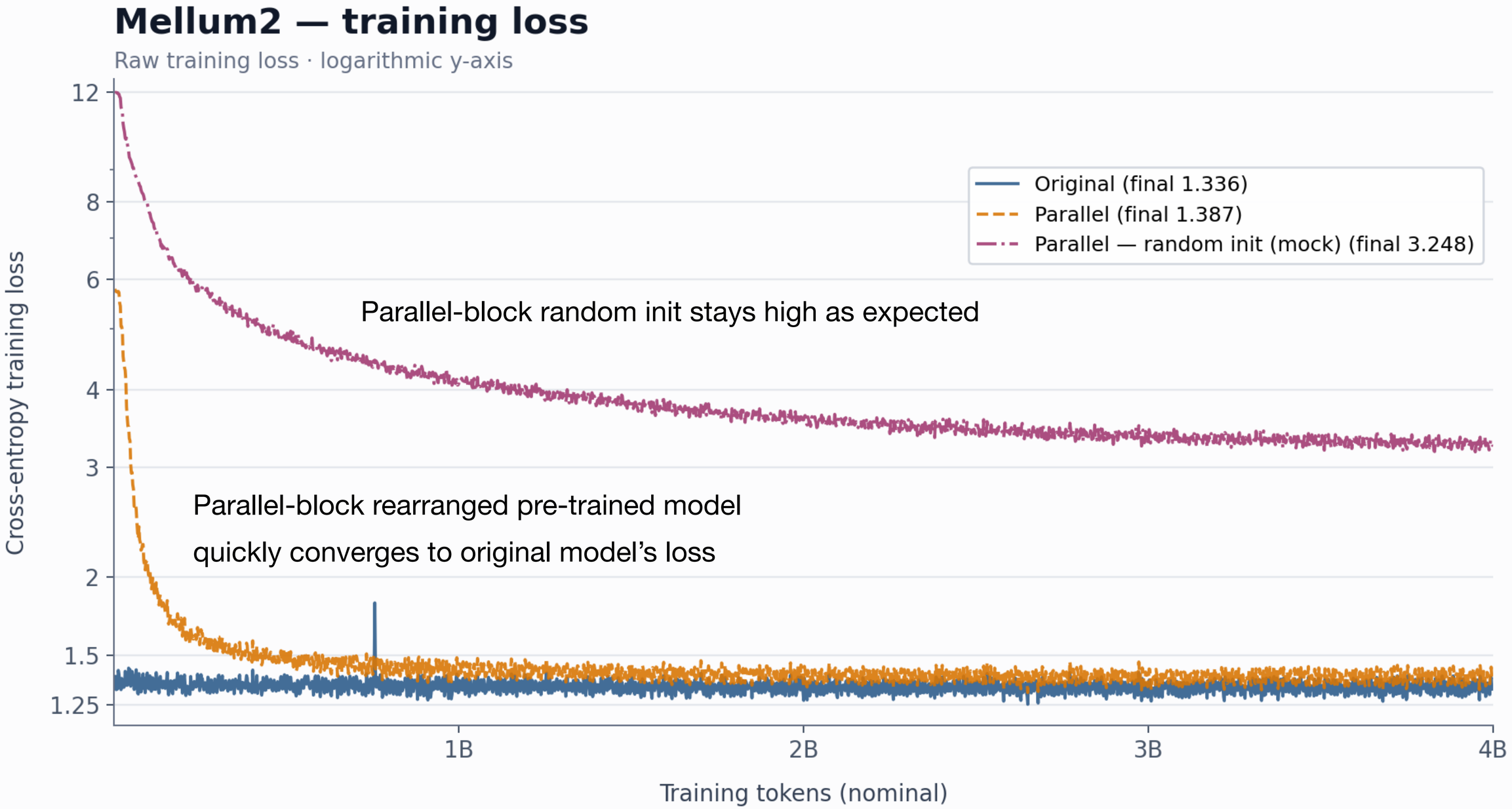
Converted model: three parallel SWA/MoE, one sequential Dense/MoE



Parallel blocks 1–3: $x_{i+1} = x_i + A_i\big(N_i^a(x_i)\big) + M_i\big(N_i^m(x_i)\big)$. **Block 4:** original sequential computation.

Residual paths are drawn explicitly; N_i^a and N_i^m are the original attention and MoE norms. Repeat this four-block cycle as needed.

3 parallel SWA/MoE blocks : 1 seq dense attn/MoE based on Mellum2 SWA/dense arch

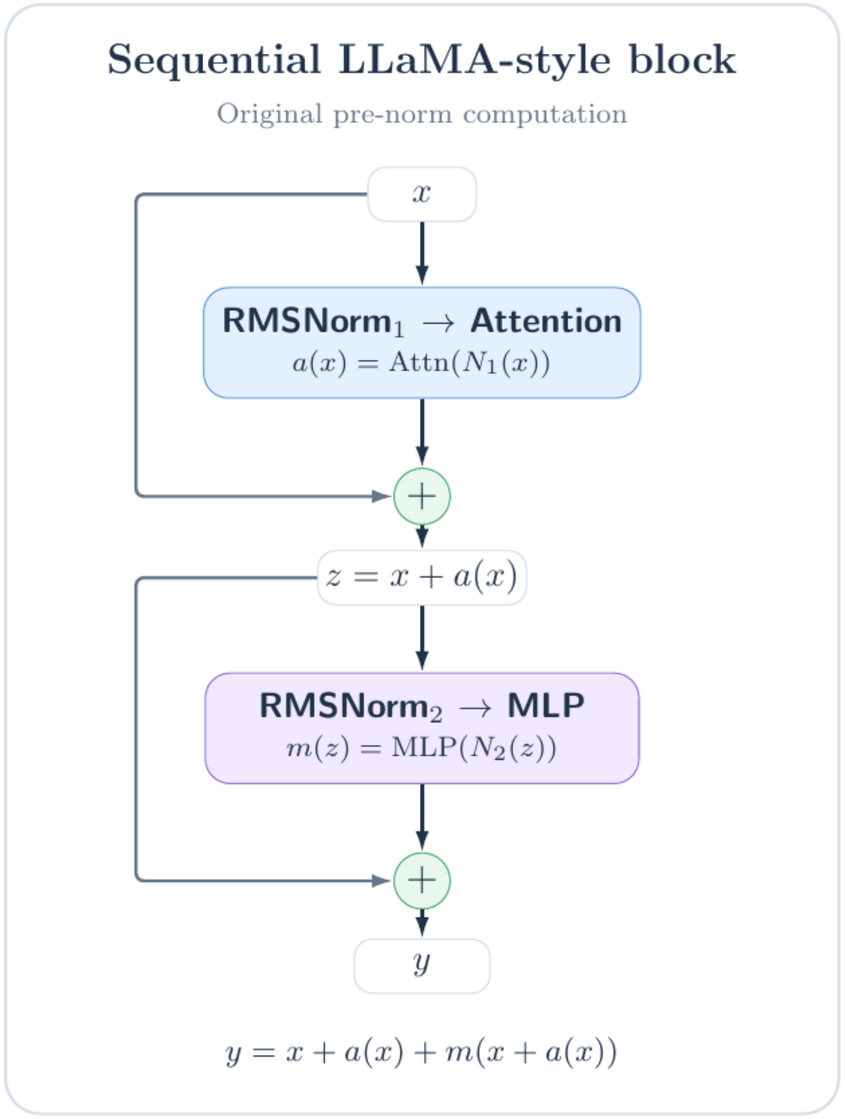


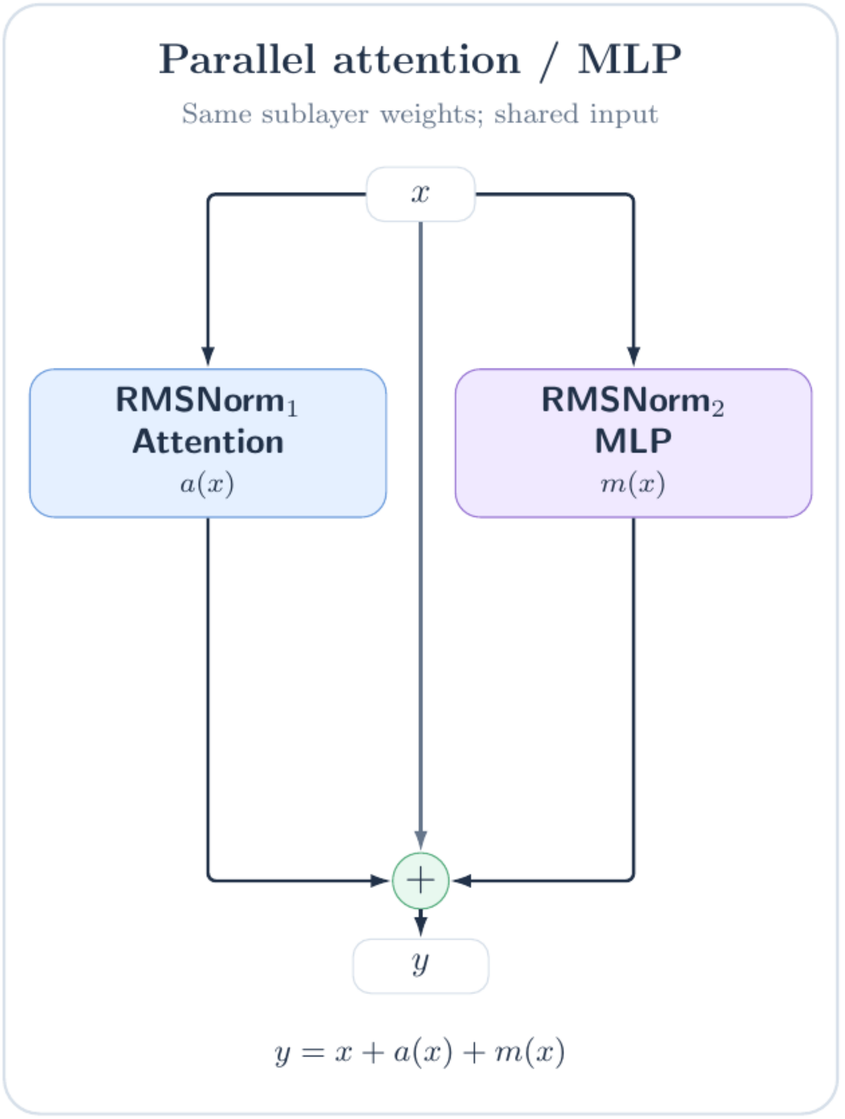
Mellum2 parallel arch downstream evals

Training budget 4B tokens

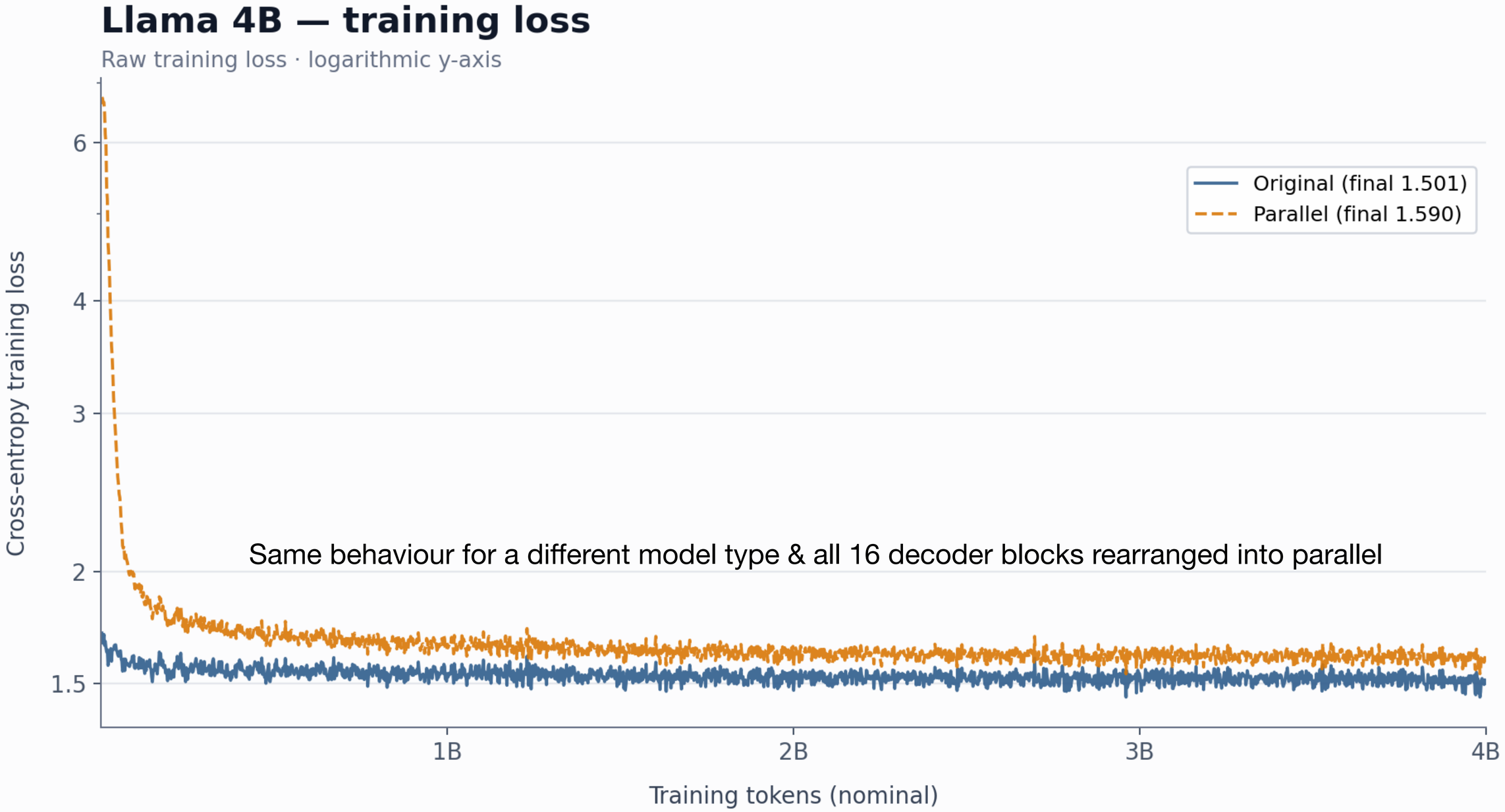
Model	Mean	BBH	GPQA	MuSR	BoolQ	PIQA	HellaSwag	WinoGrande	ARC Easy	ARC Challenge	MMLU Cont.
Original	61.57%	51.92%	35.23%	45.37%	83.39%	79.33%	74.44%	66.85%	79.38%	54.52%	45.26%
Parallel (rearrange original)	56.32%	40.01%	30.37%	40.74%	80.06%	77.86%	70.67%	63.93%	71.51%	45.99%	42.05%
Parallel (random init)	34.03%	31.21%	24.33%	33.60%	42.20%	54.13%	25.74%	49.96%	33.33%	22.53%	23.30%

All 16 decoder blocks in Llama 4B here are rearranged into parallel attn/MLP





All 16 decoder blocks in Llama 4B here are rearranged into parallel attn/MLP



Llama 4B 16L parallel arch downstream evals

Training budget 4B tokens

Model	**Mean**	BBH	GPQA	MuSR	BoolQ	PIQA	HellaSwag	WinoGrande	ARC Easy	ARC Challenge	MMLU
Original	**55.05%**	37.91%	26.68%	41.80%	77.92%	76.55%	69.15%	68.27%	69.53%	44.54%	38.12%
Parallel (rearrange original)	**51.15%**	34.59%	25.42%	38.23%	71.56%	73.67%	63.24%	63.85%	65.36%	40.61%	35.00%

Modification: shifted overlap of parallel blocks

Shifted attention–MLP overlap

Four decoder blocks shown; each attention and MLP keeps its own pre-norm and original weights

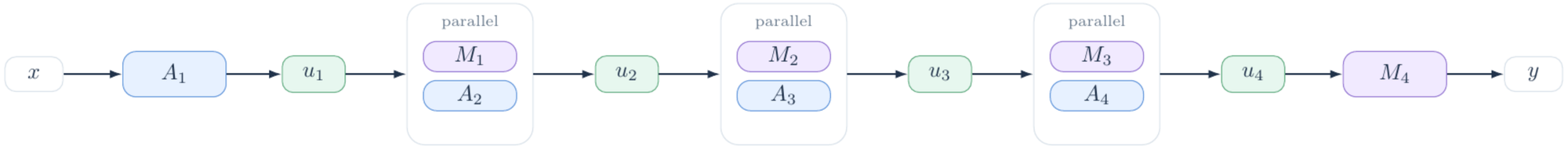
Original sequential order

Attention output enters its MLP; MLP output enters the next attention



Rearranged order

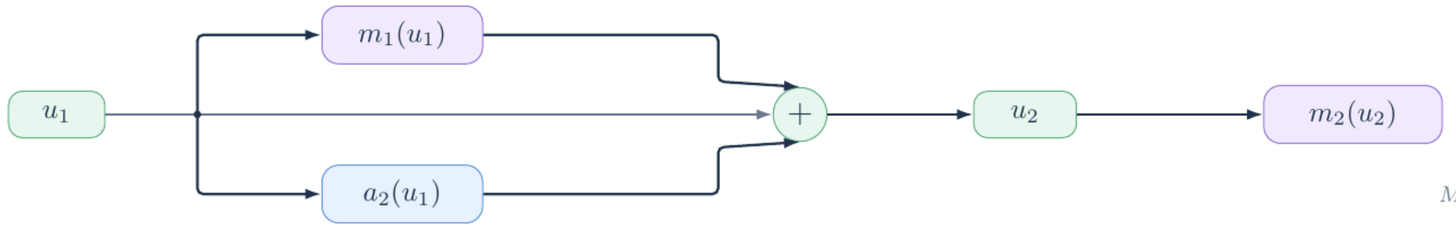
Parentheses are simultaneous operations on the same residual state



$$u_1 = x + a_1(x), \qquad u_{i+1} = u_i + m_i(u_i) + a_{i+1}(u_i), \qquad y = u_4 + m_4(u_4)$$

Inside the first parallel pair

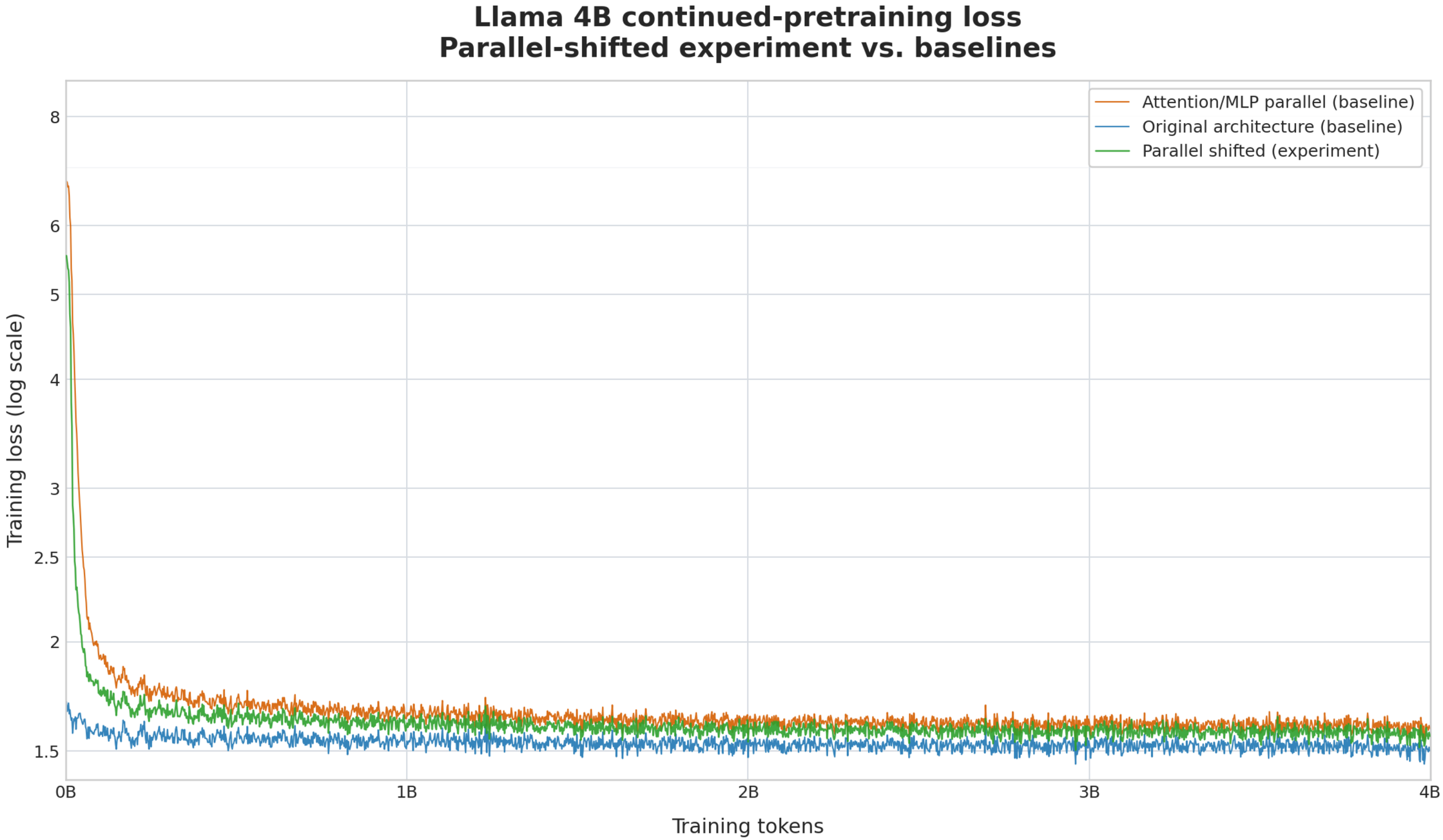
The same pattern repeats for $M_2 \parallel A_3$ and $M_3 \parallel A_4$



M_2 sees A_2's update

$$u_2 = u_1 + m_1(u_1) + a_2(u_1)$$

Modification: shifted overlap of parallel blocks (Llama 4B)

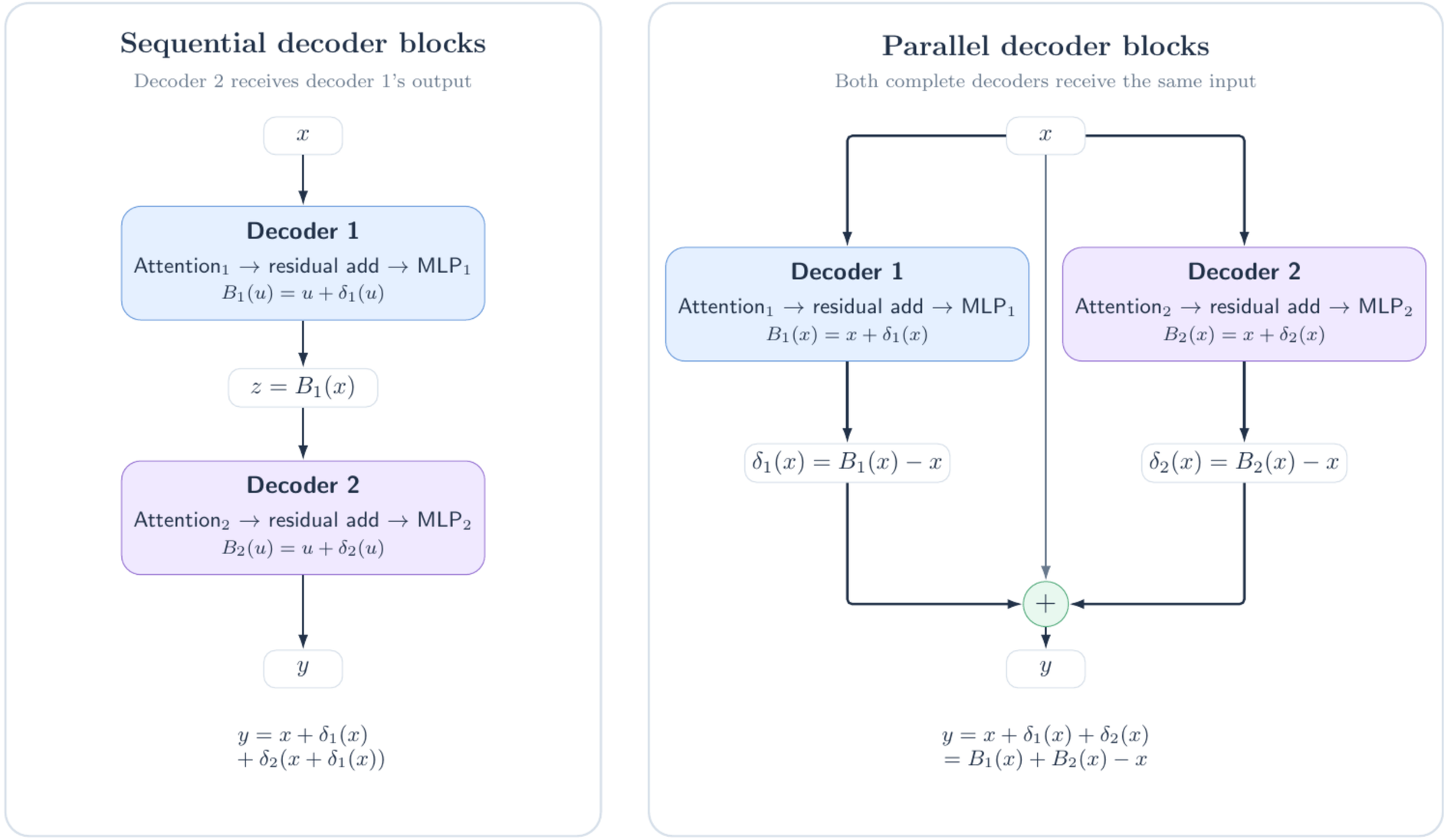


Llama 4B 16L parallel arch downstream evals

Training budget 4B tokens

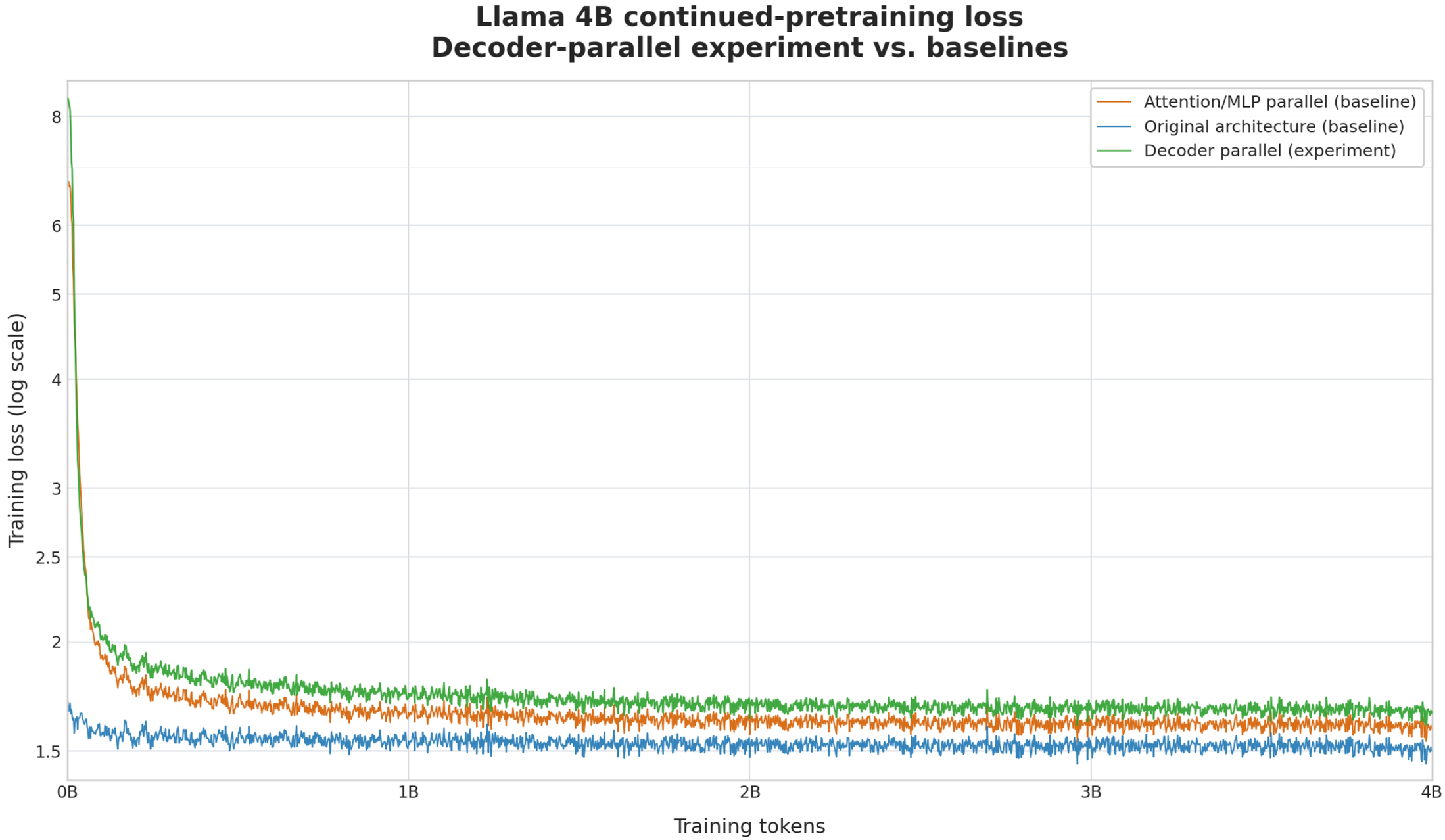
Model	**Mean**	BBH	GPQA	MuSR	BoolQ	PIQA	HellaSwag	WinoGrande	ARC Easy	ARC Challenge	MMLU
Original	**55.05%**	37.91%	26.68%	41.80%	77.92%	76.55%	69.15%	68.27%	69.53%	44.54%	38.12%
Parallel (rearrange original)	**51.15%**	34.59%	25.42%	38.23%	71.56%	73.67%	63.24%	63.85%	65.36%	40.61%	35.00%
Parallel-shifted	**51.61%**	34.7%	22.99%	41.01%	72.11%	72.85%	65.02%	64.48%	67.63%	40.7%	34.59%

More aggressive depth reduction: parallel decoder blocks rearrangement



Inside each decoder: $\delta_i(u) = a_i(u) + m_i(u + a_i(u))$; attention and MLP stay sequential.

More aggressive depth reduction: parallel decoder blocks rearrangement



Llama 4B 16L parallel arch downstream evals

Training budget 4B tokens

Model	**Mean**	BBH	GPQA	MuSR	BoolQ	PIQA	HellaSwag	WinoGrande	ARC Easy	ARC Challenge	MMLU
Original	**55.05%**	37.91%	26.68%	41.80%	77.92%	76.55%	69.15%	68.27%	69.53%	44.54%	38.12%
Parallel attn/ MLP	**51.15%**	34.59%	25.42%	38.23%	71.56%	73.67%	63.24%	63.85%	65.36%	40.61%	35.00%
Decoder-parallel	**50.82%**	34.49%	25.00%	41.67%	67.98%	74.27%	60.16%	67.01%	64.86%	39.08%	33.66%

The change in block representation has a form (both for attn/mlp and parallel decoder)

$$x + a(x) + m(x + a(x)) \quad \longrightarrow \quad x + a(x) + m(x)$$

Hence if we wanted to introduce a correction term

$$y_t = x_t + a_t + m_t(x_t) + c_\theta(x_t, a_t)$$

where *c(x, a)* implements both token-wise and feature-wise mixing:

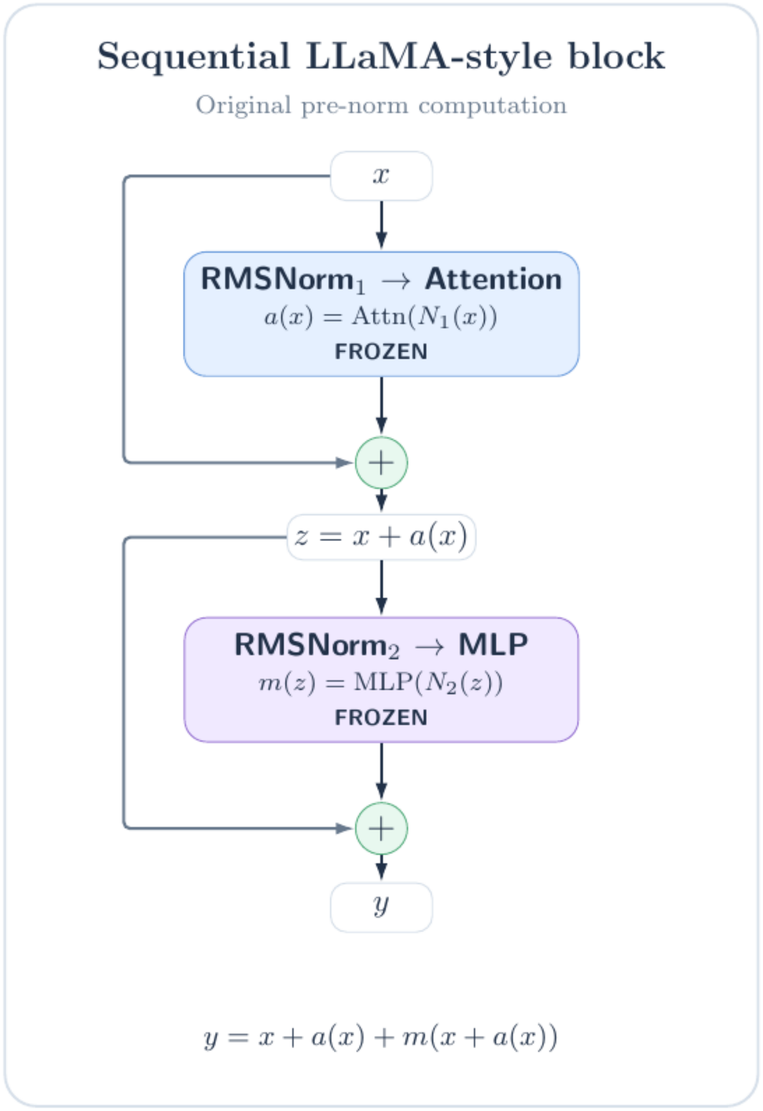
$$c_\theta(x_t, a_t) \;\approx\; m_t(x_t + a_t) - m_t(x_t)$$

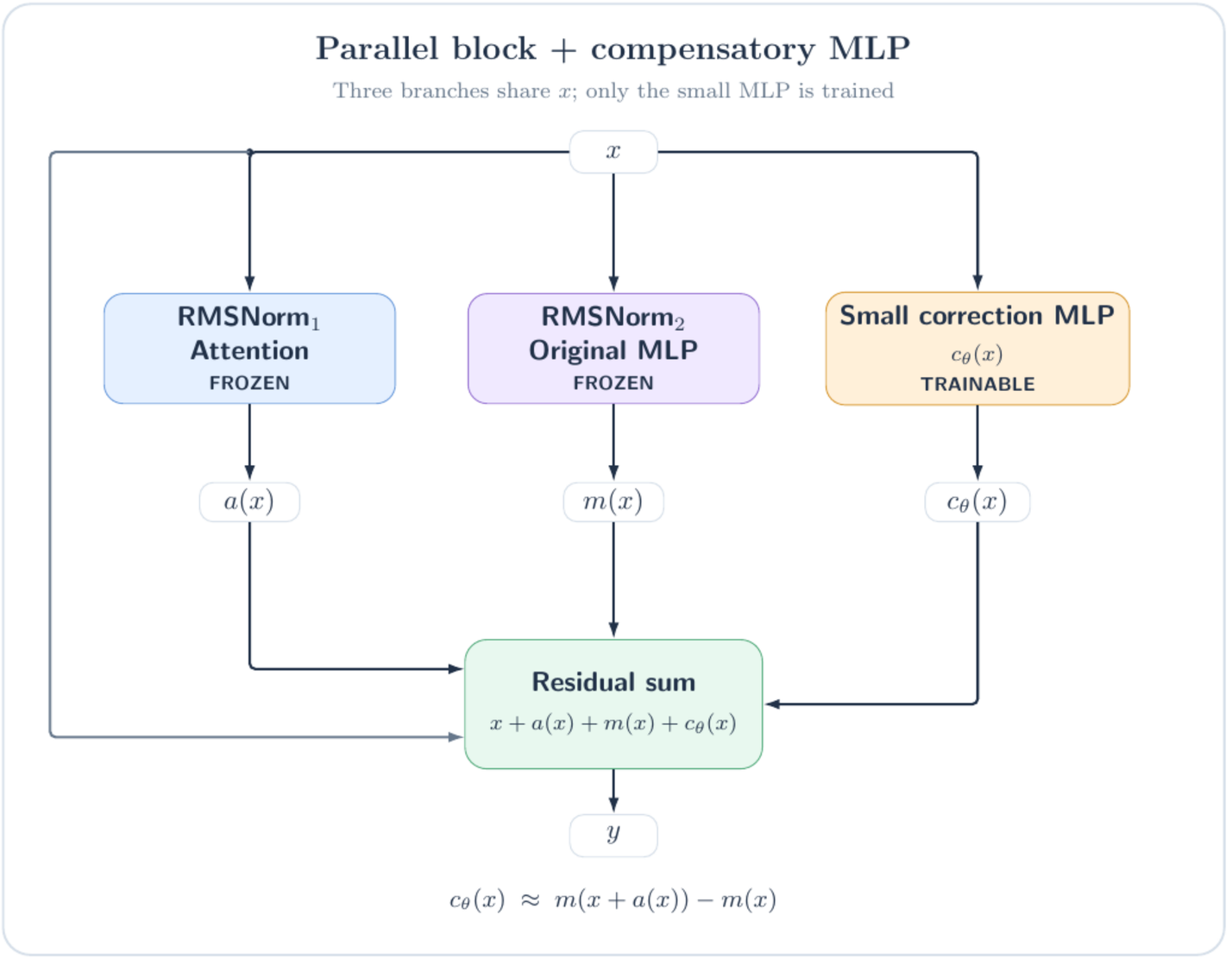
Alternatively, under the approximation *a(x)* << *x* (update to residual stream is small)

$$m(x + a(x)) \approx m(x) + J_m(x)\, a(x)$$

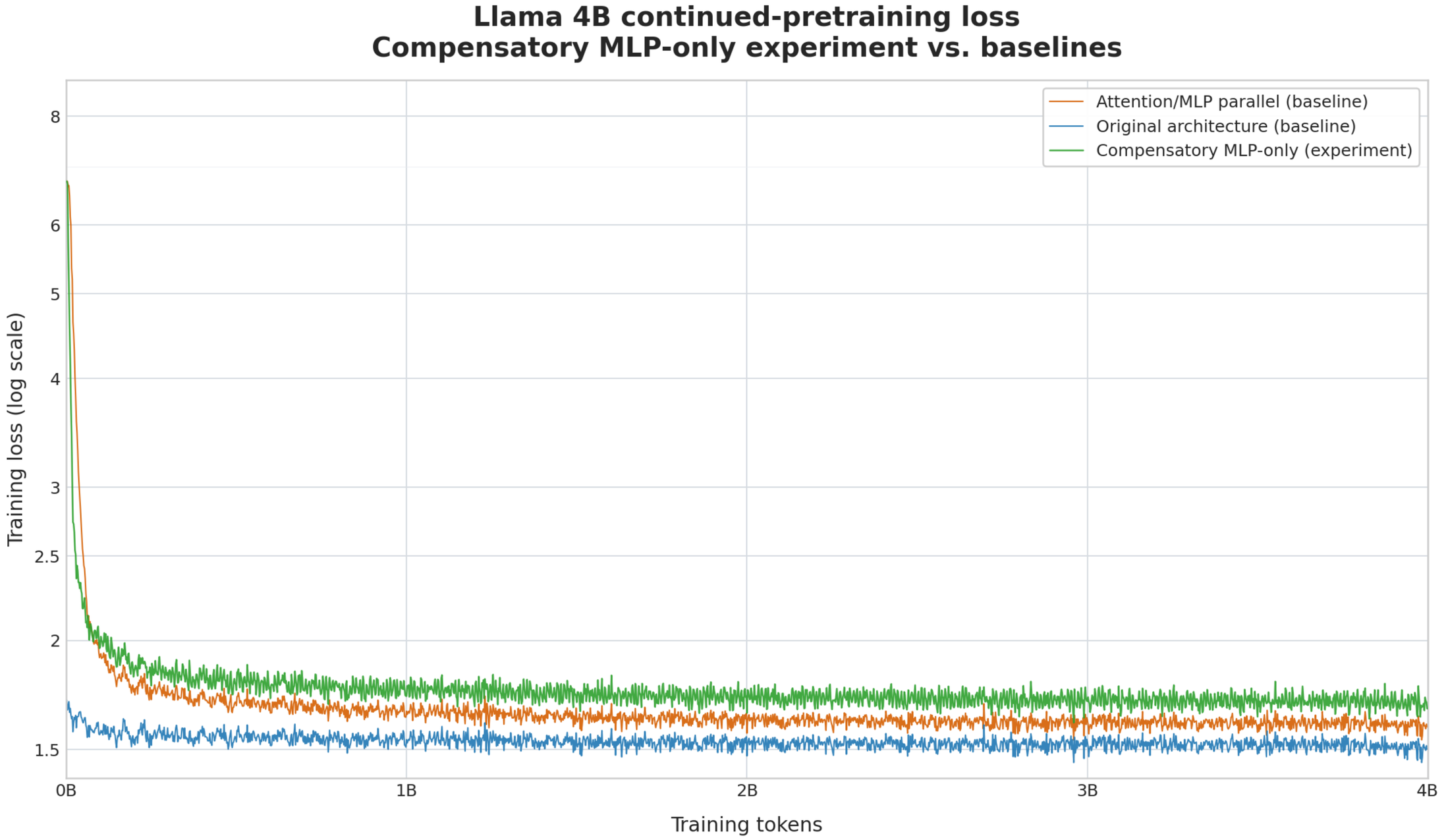
Hence, we can add a learnable gate to attn/first decoder that approximates he Jacobian J_m(x)

Freeze all original weights, add a lightweight correction branch and train its parameters





Compensatory MLP has expansion factor = 1, +0.5B total model params



Llama 4B 16L parallel arch downstream evals

Training budget 4B tokens

Model	**Mean**	BBH	GPQA	MuSR	BoolQ	PIQA	HellaSwag	WinoGrande	ARC Easy	ARC Challenge	MMLU
Original	**55.05%**	37.91%	26.68%	41.80%	77.92%	76.55%	69.15%	68.27%	69.53%	44.54%	38.12%
Parallel attn/ MLP	**51.15%**	34.59%	25.42%	38.23%	71.56%	73.67%	63.24%	63.85%	65.36%	40.61%	35.00%
Train only compensatory MLP	**55.2%**	38.21%	28.1%	46.03%	76.64%	76.01%	66.97%	68.51%	71.00%	43.52%	37.05%